Sampling Distributions

I. Expected Values (Again)

II. Estimators of the Population Mean

A. Properties

III. What is a Sampling Distribution

IV. The Central Limit Theorem

V. Other Sampling Distributions

Fundamentals of Clinical Trials

I. Elements of Clinical Trials

II. What do the Data Show?

Outline

- independentSamplesTTest()
- oneSampleTTest()
- posthocPairwiseT()
  - Bonferroni correction
  - Holm correction

Summary of Some Statistical Methods to Test Hypotheses

Scale of measurement	Type of experiment: Two treatment groups consisting of different individuals	Three or more treatment groups consisting of different individuals	Before and after a single treatment in the same individuals	Multiple treatments in the same individuals	Association between two variables
Interval (and drawn from normally distributed populations*)	Unpaired *t* test (Chapter 4)	Analysis of variance (Chapter 3)	Paired *t* test (Chapter 9)	Repeated-measures analysis of variance (Chapter 9)	Linear regression, Pearson product-moment correlation, or Bland-Altman analysis (Chapter 8)
	z Test				
Nominal	Chi-square analysis-of-contingency table (Chapter 5)	Chi-square analysis-of-contingency table (Chapter 5)	McNemar's test (Chapter 9)	Cochrane Q†	Relative risk or odds ratio (Chapter 5)
Ordinal‡	Mann-Whitney rank-sum test (Chapter 10)	Kruskal-Wallis statistic (Chapter 10)	Wilcoxon signed-rank test (Chapter 10)	Friedman statistic (Chapter 10)	Spearman rank correlation (Chapter 8)
Survival time	Log-rank test or Gehan's test (Chapter 11)				

*If the assumption of normally distributed populations is not met, rank the observations and use the methods for data measured on an ordinal scale.

†Not covered in this text.

‡Or interval data that are not necessarily normally distributed.

Motivation

- Understanding sampling distributions helps us move from descriptive statistics for a particular sample group to inferential statistics about an entire population (or multiple populations)

- Gives us an idea of how well a single group represents the entire population

I. Expected Value

Let X denote a continuous RV with $X \sim f(x)$

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx = \mu$$

Define a function of X as $h(X)$

$$E[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) dx$$

e.g.: $h(X) = (X - \mu)^2$ $\quad E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx = \sigma^2$

Mean and Variance of Linear Combinations of RVs

$$Y = c_1 X_1 + \cdots + c_m X_m$$

$$\begin{cases} X_i \sim f_i(x_i) \\ X_i, X_j \text{ independent } i \neq j \\ c_i \text{ constants} \end{cases}$$

$$E[Y] = E[c_1 X_1 + \cdots + c_m X_m] = E[c_1 X_1] + \cdots + E[c_m X_m]$$

$$= \sum_{i=1}^{m} c_i E[X_i]$$

$$E\left[\left(Y - E(Y)\right)^2\right] = \sum_{i=1}^{m} c_i^2 E\left[\left(X_i - E[X_i]\right)^2\right] = \sum_{i=1}^{m} c_i^2 Var(X_i)$$

Note: if $X_i \sim N\left(\mu_{X_i}, \sigma^2_{X_i}\right)$, then $Y \sim N\left(\mu_Y, \sigma^2_Y\right)$

$$\begin{cases} \mu_Y = \sum_{i=1}^{m} c_i \mu_{X_i} \\ \sigma^2_Y = \sum_{i=1}^{m} c_i^2 \sigma^2_{X_i} \end{cases}$$

II. Estimators of the Population Mean

Birthweight in oz.

$$X \sim N(\mu, \sigma^2)$$

How to <u>estimate</u>?

$$\mu \ , \ \sigma^2$$

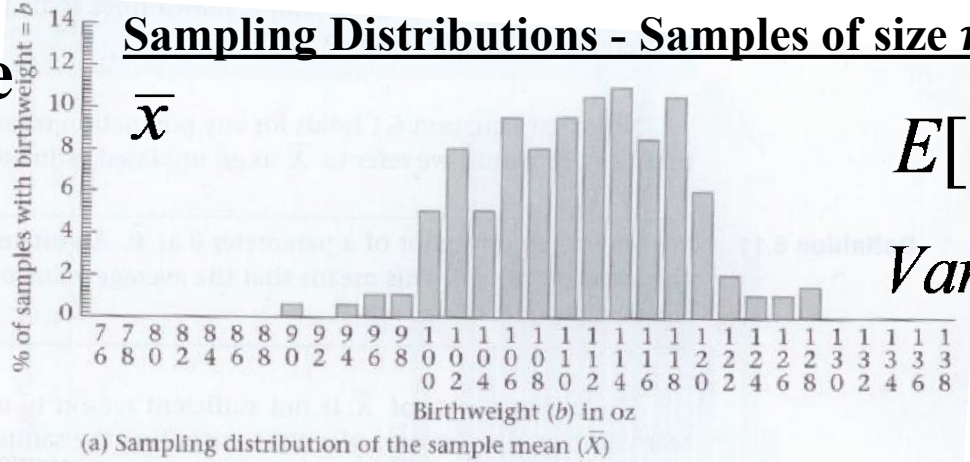
Three diff. estimators for μ
1) Sample mean
2) Sample median
3) Avg. of min & max

<u>Sampling Distributions - Samples of size $n = 10$</u>

$\bar{x}$

$$E[\bar{x}] = \mu$$

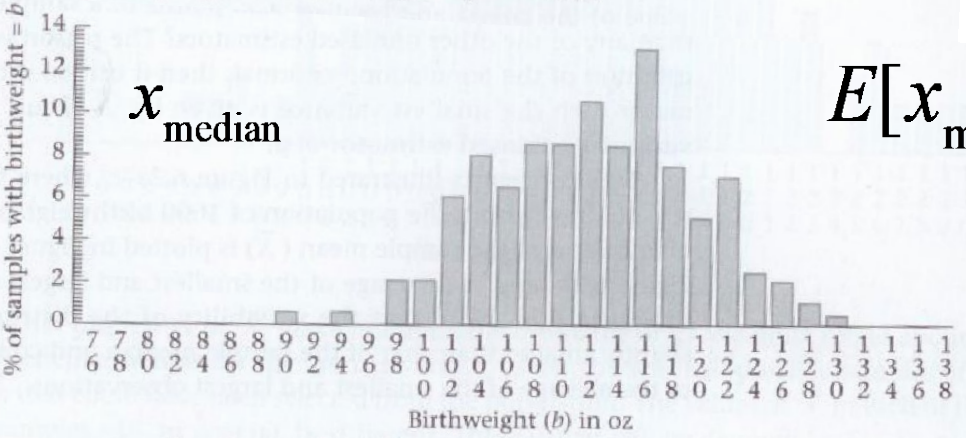
$$Var(\bar{x}) = \frac{\sigma^2}{n}$$



(a) Sampling distribution of the sample mean ($\bar{X}$)

x_{median}

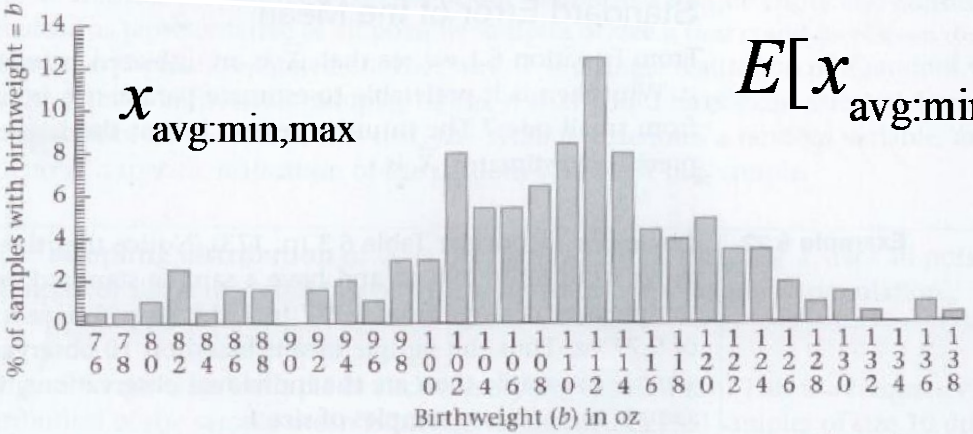
$$E[x_{median}] = \mu$$



(b) Sampling distribution of the sample median

$x_{avg:min,max}$

$$E[x_{avg:min,max}] = \mu$$



(c) Sampling distribution of the average of the smallest and largest observations

A. Properties of Sample Mean

$$E[\bar{x}] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} E[X] = \frac{1}{n}\sum_{i=1}^{n} \mu = \frac{1}{n}(n\mu) = \mu$$

$$Var(\bar{x}) = \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n} \qquad \frac{\sigma}{\sqrt{n}} \leftarrow SEM$$

$$Var(x_{\text{median}}) > Var(\bar{x})$$

$$Var(x_{\text{avg:min,max}}) > Var(\bar{x})$$

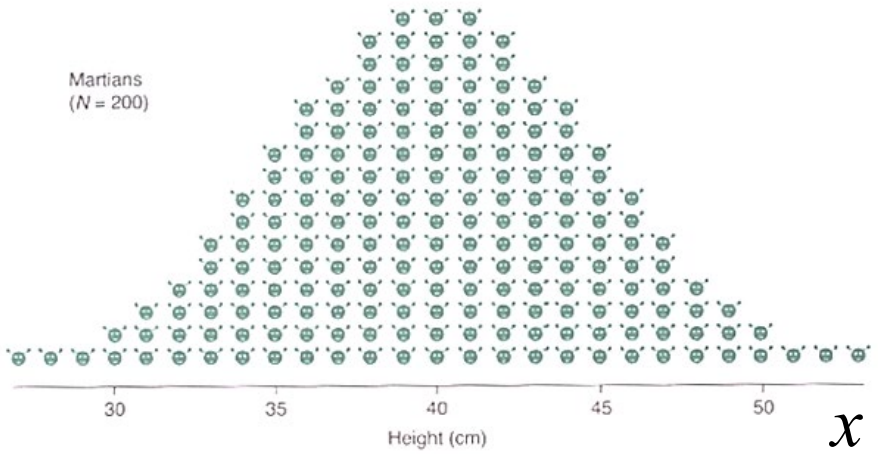
Sample Mean $\bar{x}$ is an unbiased, minimum-variance estimator of μ

III. What is a Sampling Distribution?

e.g,
F and t

- Any statistic calculated from sample data is another random variable, with its own distribution
- For example, the sample mean has a probability distribution: how the statistic varies in value across all sample groups of size n that might be selected
- If you choose 25 sample groups of size n, you will get 25 different sample means
- Each of these sample means is an estimate of the population mean and the distribution of all of the possible values of the sample mean is called a **sampling distribution**

Sampling Distribution of $\bar{x}$ (the Sample Mean)



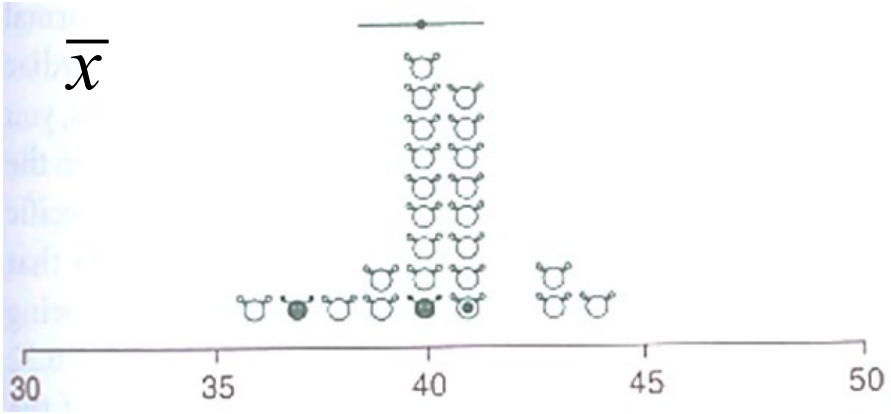
Let RV X represent height

$$X \sim f(x)$$

(original random variable)

$$E[X] \equiv \mu$$

$$E[(X - E[X])^2] = Var(X) = \sigma^2$$



Sample means for samples of size 10

Define a new RV, $\bar{x}$, as the sample mean for sample of size n

$\bar{x}$ is a function of the original RV and has its own probability distribution (only partially shown here)

IV. The Central Limit Theorem

$X \sim f(x)$ with μ, σ^2

for large n, $\bar{x} \dot{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$

Approximately distributed

Note:

if $X \sim N(\mu, \sigma^2)$, then $\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ for all values of n

Figure 4.1 *Uniform Population*

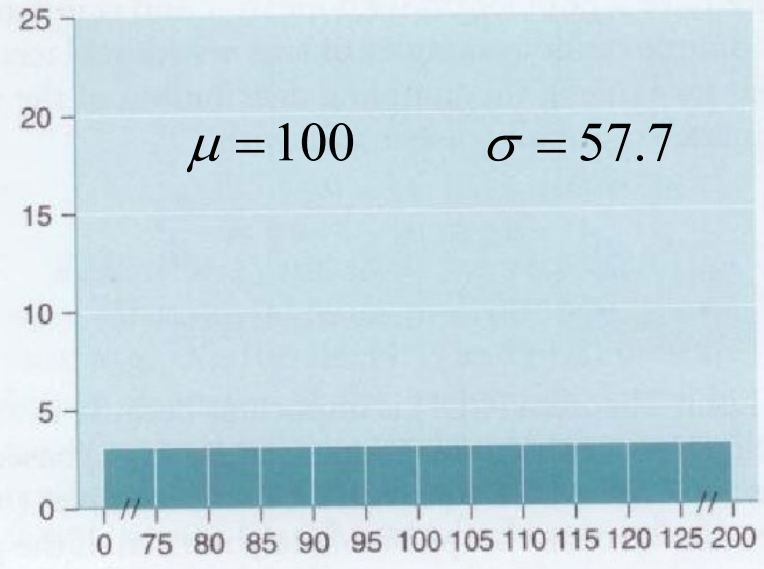
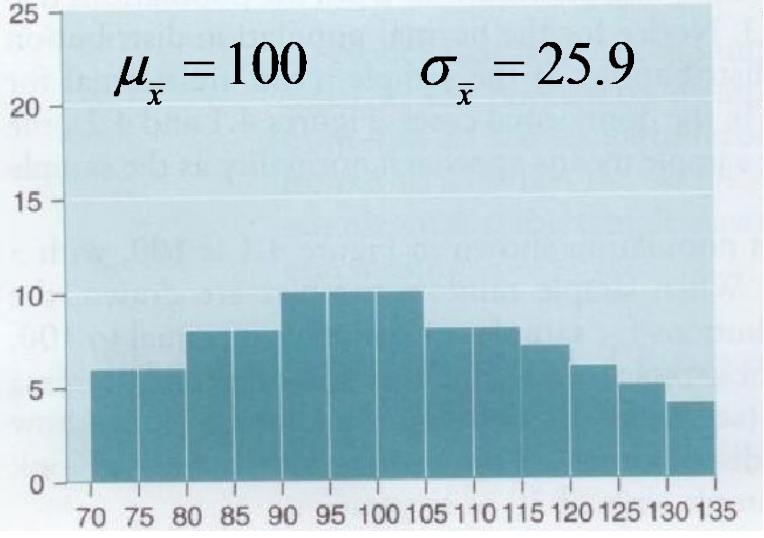
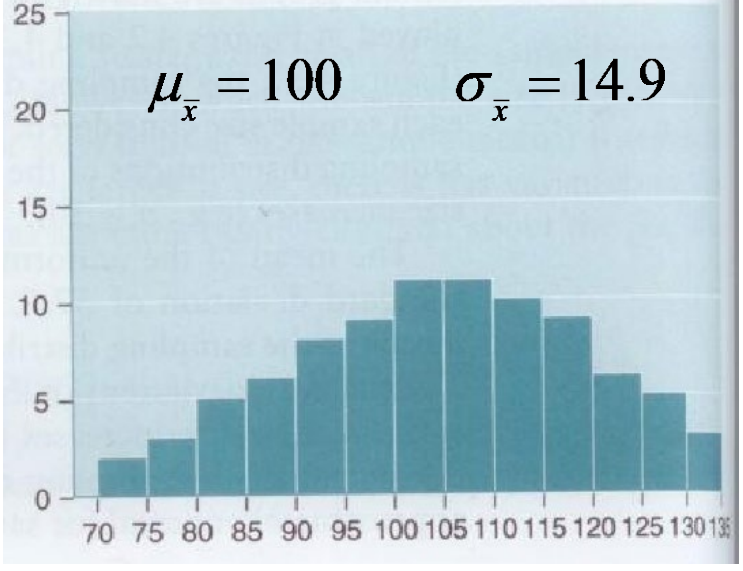


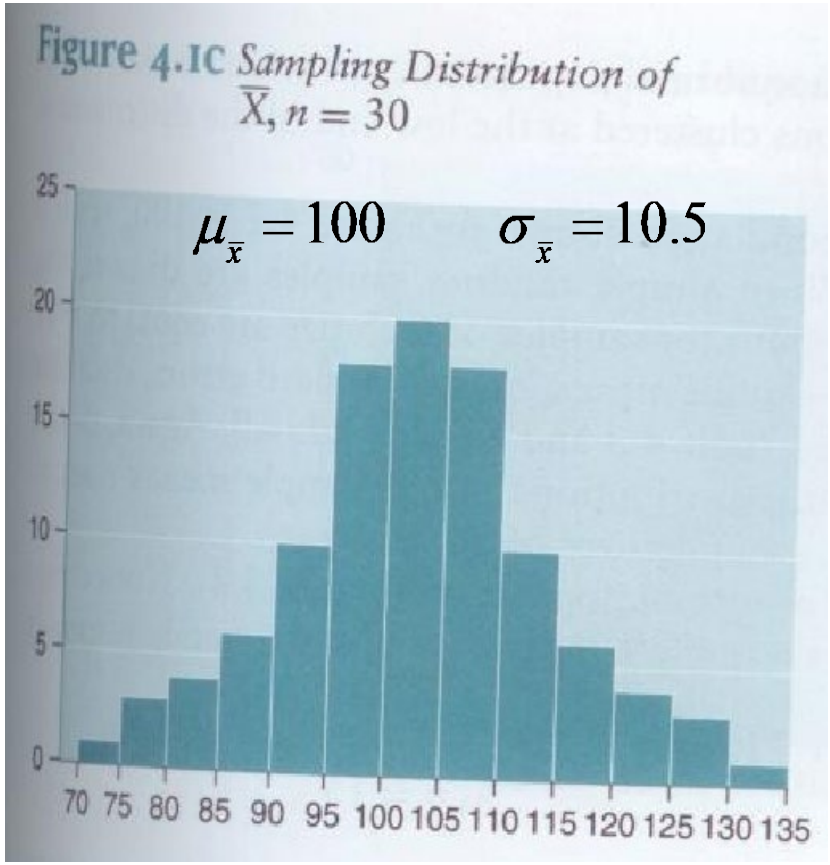
Figure 4.1a *Sampling Distribution of* $\overline{X}$, $n = 5$

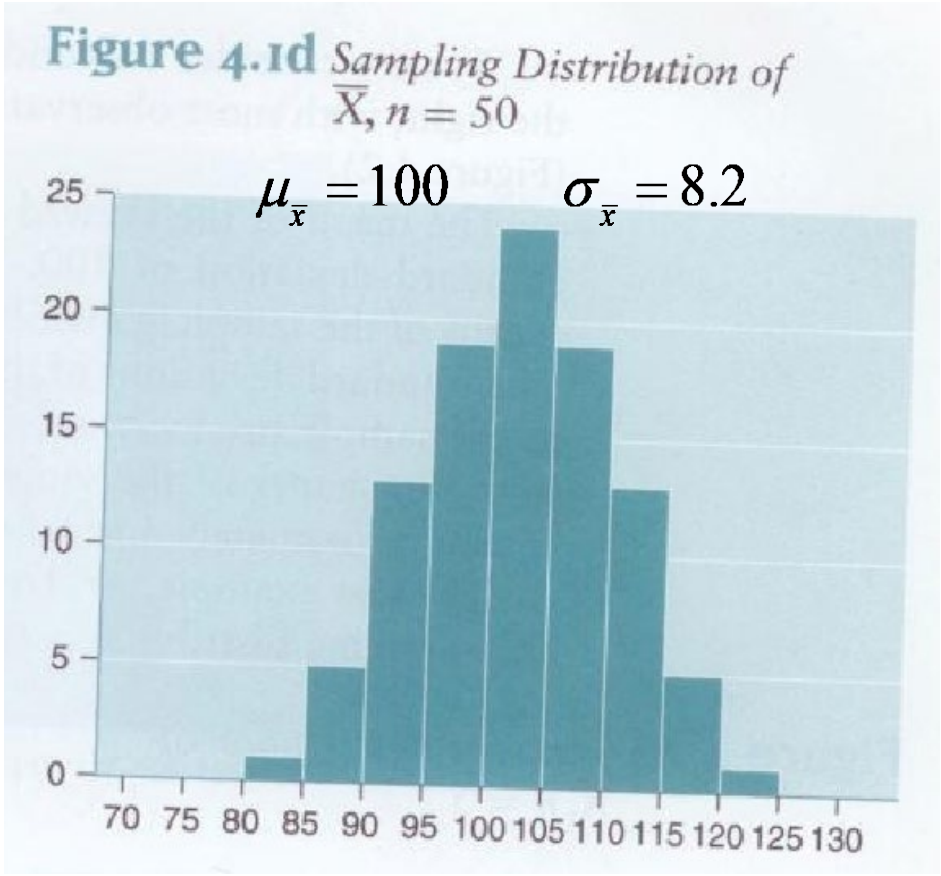


$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

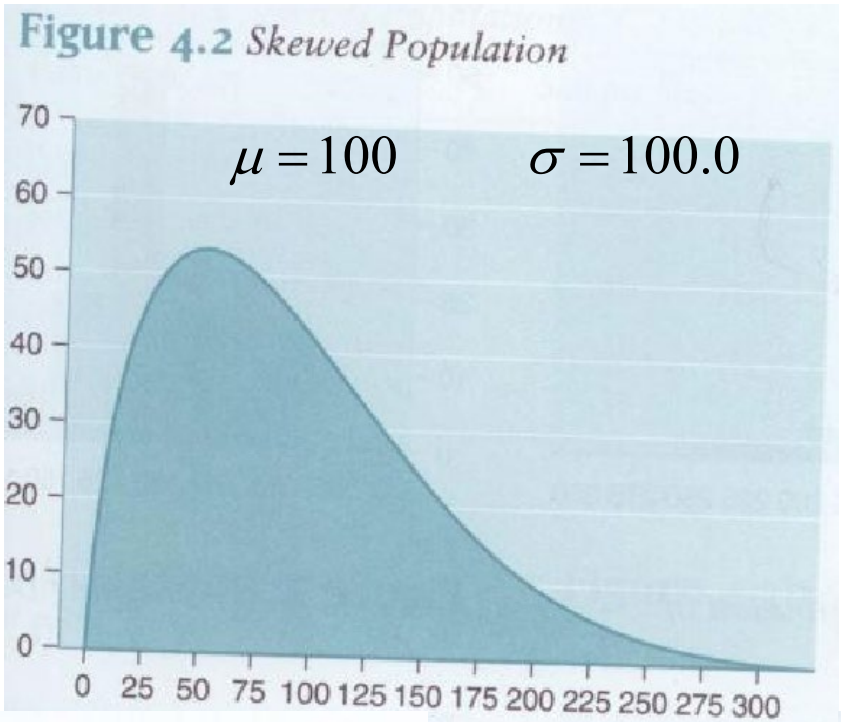
Figure 4.1b *Sampling Distribution of* $\overline{X}$, $n = 15$

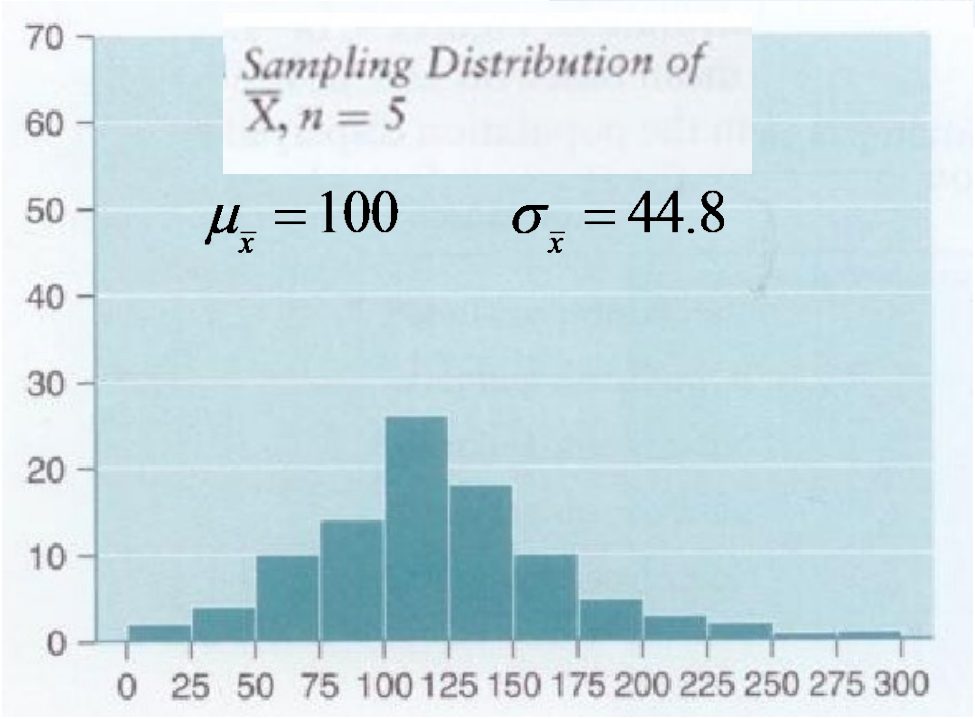






$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$


Figure 4.2 Skewed Population
$\mu = 100$
$\sigma = 100.0$
70
60
50
40
30
20
10
0
0 25 50 75 100 125 150 175 200 225 250 275 300


Sampling Distribution of $\overline{X}$, $n = 5$
$\mu_{\bar{x}} = 100$
$\sigma_{\bar{x}} = 44.8$
70
60
50
40
30
20
10
0
0 25 50 75 100 125 150 175 200 225 250 275 300

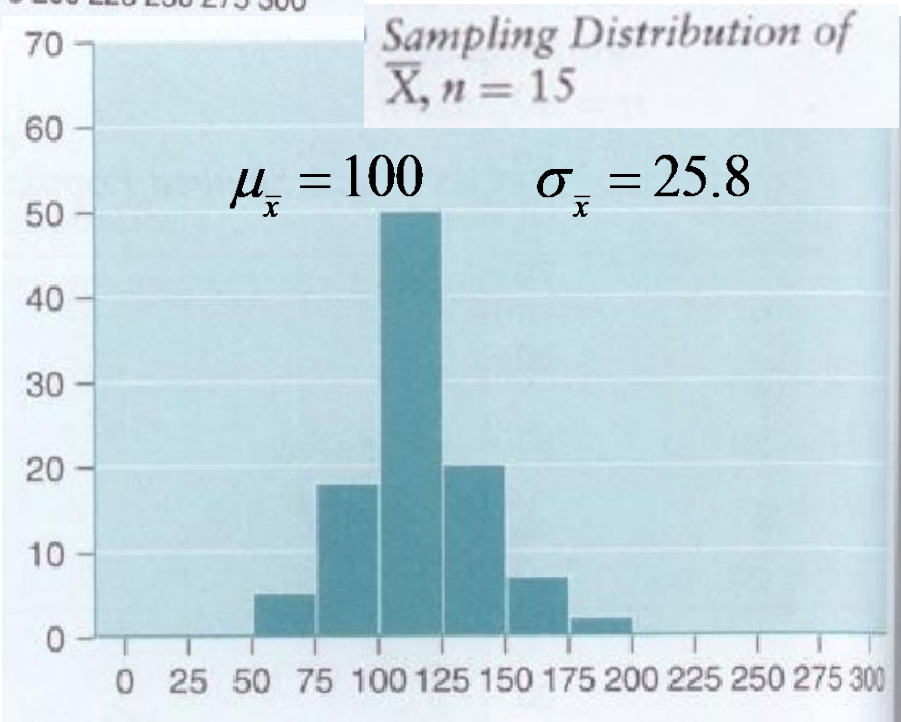

Sampling Distribution of $\overline{X}$, $n = 15$
$\mu_{\bar{x}} = 100$
$\sigma_{\bar{x}} = 25.8$
70
60
50
40
30
20
10
0
0 25 50 75 100 125 150 175 200 225 250 275 300

Figure 4.2c *Sampling Distribution of* $\overline{X}$, $n = 30$

$\mu_{\bar{x}} = 100$ $\sigma_{\bar{x}} = 18.2$

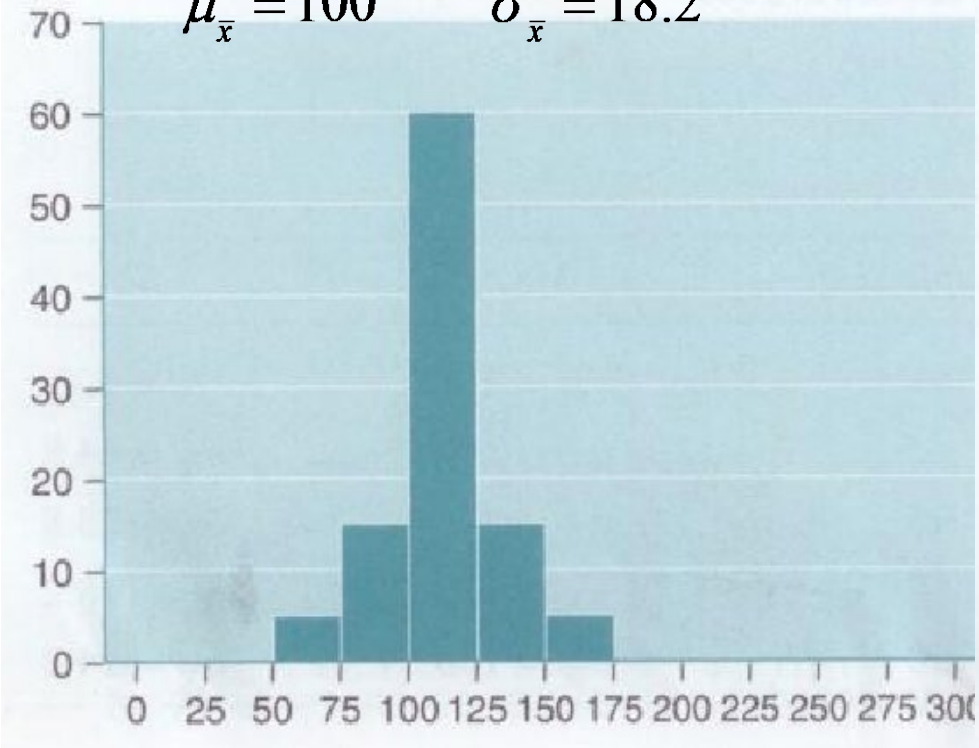
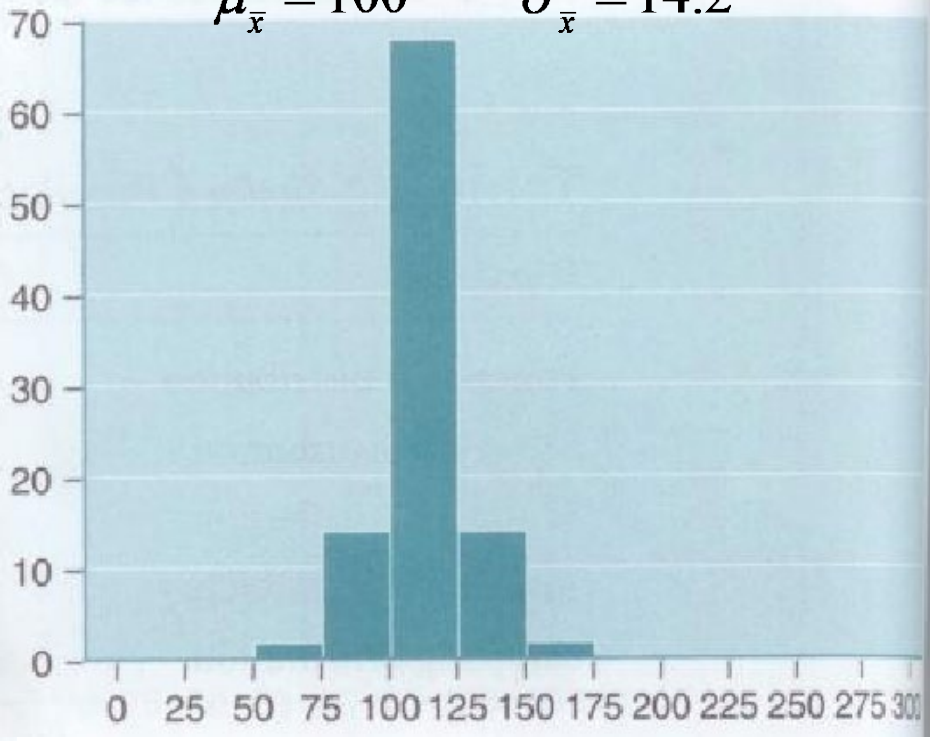


Figure 4.2d *Sampling Distribution of* $\overline{X}$, $n = 50$

$\mu_{\bar{x}} = 100$ $\sigma_{\bar{x}} = 14.2$



$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

V. Other Sampling Distributions

$$X \sim N(\mu, \sigma^2)$$

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Define RV Z

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \qquad Z \sim N(0, 1)$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$$

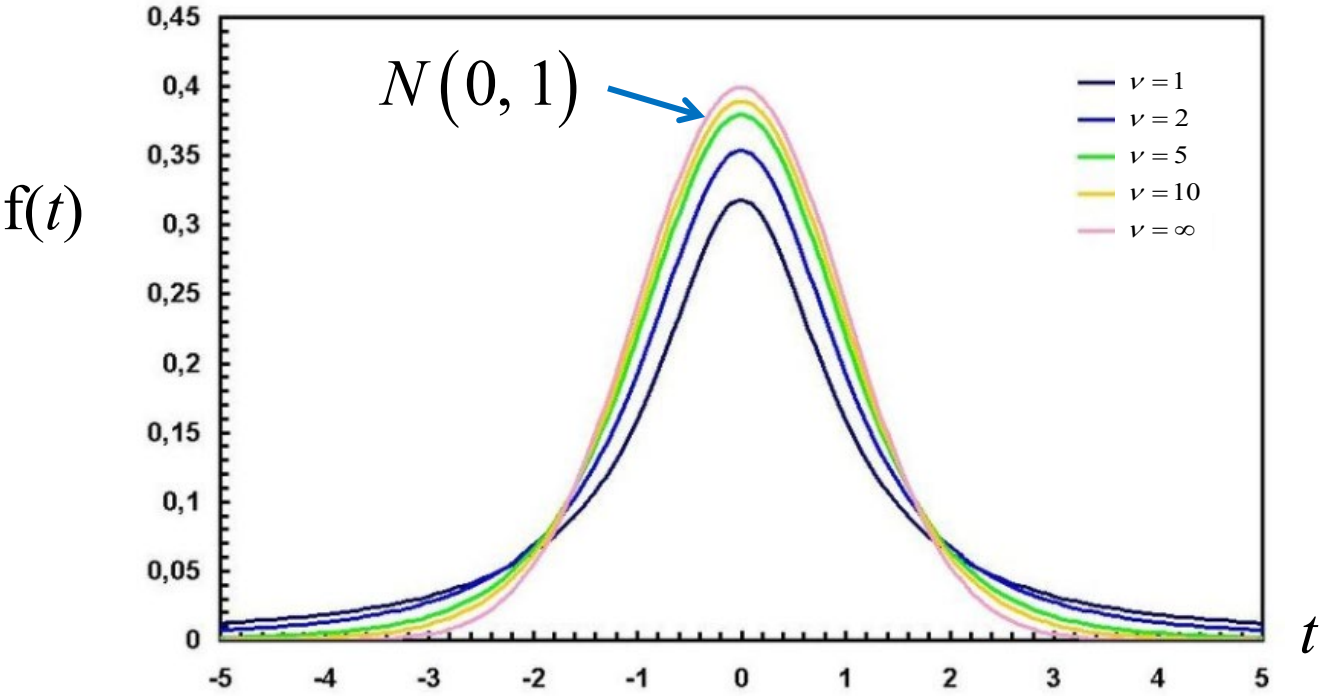
In an experiment σ is unknown but can be estimated by s

$$\frac{\bar{x} - \mu}{s / \sqrt{n}}$$

But this is not Normally Distributed

$$\frac{\bar{x} - \mu}{s / \sqrt{n}} \sim t(n-1)$$

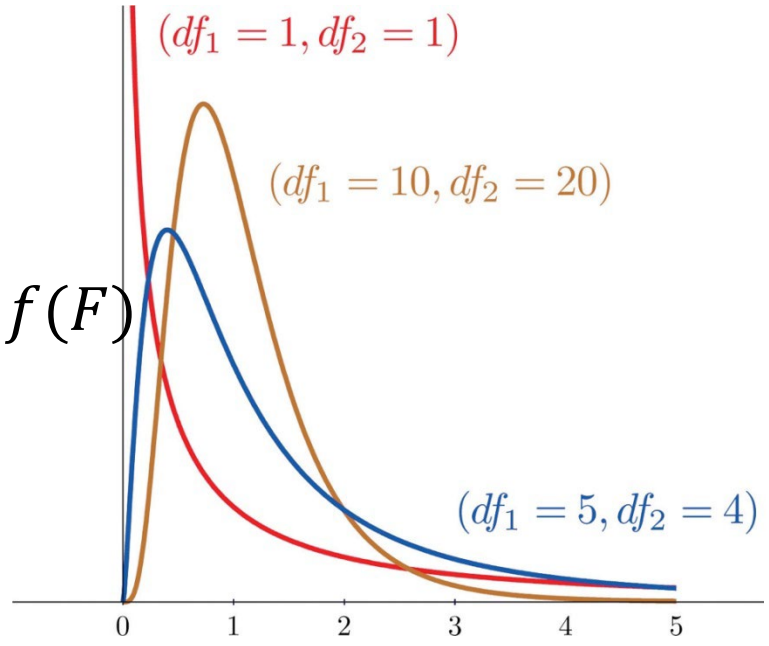
Student t Distribution p.d.f.s $t(\nu)$



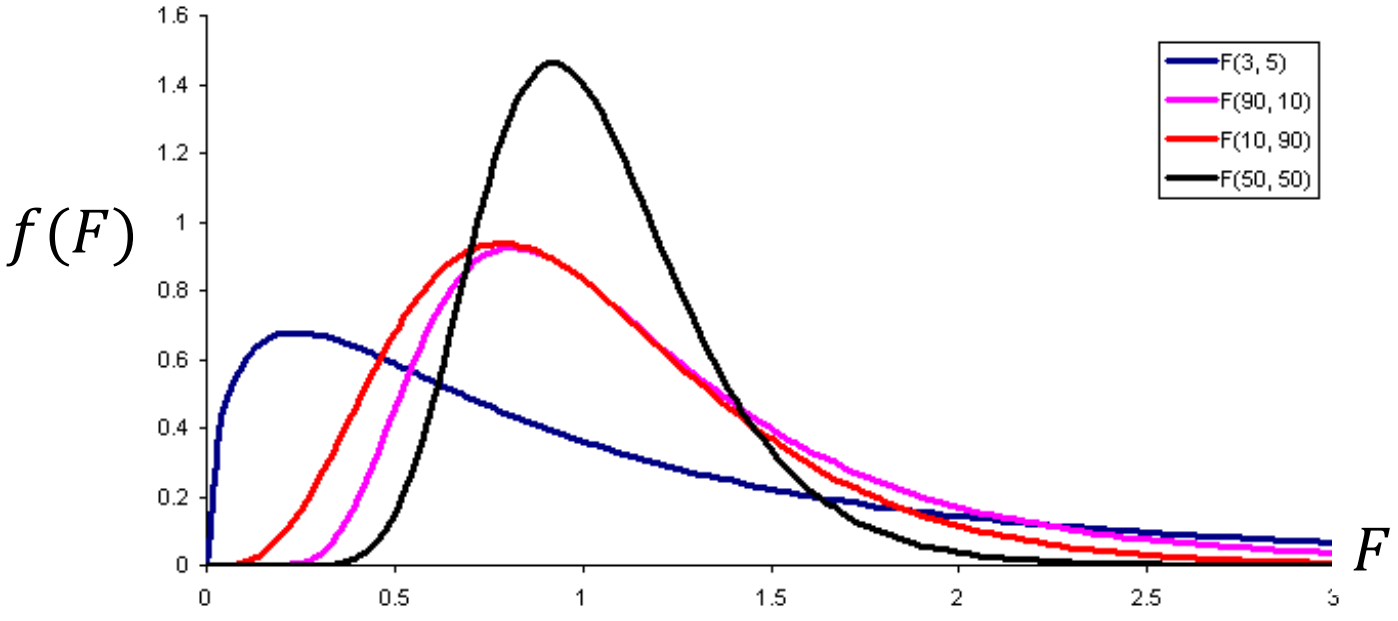
$$f(t) = \frac{\left(1 + t^2/\nu\right)^{-(\nu+1)/2}}{B(0.5,\, 0.5\nu)\sqrt{\nu}}$$

$$B(\alpha,\, \beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx$$

F Statistic Probability Density Functions (p.d.f.'s)



- F is a function of sample data, and is therefore another random variable
- Under null hypothesis of no difference, F statistic has multiple p.d.f.'s
- Probability density functions for F depend on degrees of freedom of numerator and denominator, ν_n and ν_d, calculated from number of groups and number of subjects per group: $F(\nu_n, \nu_d)$



A Comment on Degrees of Freedom and Critical Values for F

- Something you may notice from looking at the F table: increasing the degrees of freedom produces smaller critical F values

- Greater degrees of freedom (more subjects per group, more groups, or both) produce F probability density functions with smaller and smaller tails, which result in smaller values of F_{critical}

- With larger numbers of subjects per group and/or larger numbers of groups, random chance is less likely to cause a large F if all subjects truly came from the same population; thus, F_{critical} is smaller for study designs with greater degrees of freedom

Fundamentals of Clinical Trials

I. Elements of Clinical Trials

Lind Experiment on Scurvy

On the 20th May 1747, I took twelve patients in the scurvy, on board the Salisbury at sea. Their cases were as similar as I could have them. They all in general had putrid gums, the spots and lassitude, with weakness of their knees. They lay together in one place, being a proper apartment for the sick in the fore-hold; and had one diet in common to all, viz. water-gruel sweetened with sugar in the morning; fresh mutton broth often times for dinner; at other times puddings, boiled biscuit with sugar etc. And for supper, barley and raisins, rice and currants, sago and wine, or the like. Two of these were ordered each a quart of cider a day. Two others took twenty-five gutts of elixir vitriol three times a day, upon an empty stomach; using a gargle strongly acidulated with it for their mouths. Two others took two spoonfuls of vinegar three times a day, upon an empty stomach: having their gruels and their other food well acidulated with it, as also the gargle

• • •



Table 1.1. Historical Events in the Development of Clinical Trials.

Date	Author	Event
1747	Lind	Experiment with untreated control group (Lind, 1753)
1799	Haygarth	Use of sham procedure (Haygarth, 1800)
1800	Waterhouse	U.S.-based smallpox trial (Waterhouse, 1800, 1802)
1863	Gull	Use of placebo treatment (Sutton, 1865)
1923	Fisher	Application of randomisation to experimentation (Fisher and MacKenzie, 1923)
1931	—	Special committee on clinical trial created by the Medical Research Council of Great Britain (Medical Research Council, 1931)
1931	Amberson	Random allocation of treatment to groups of patients (Amberson *et al.*, 1931)
1962	Kefauver and Harris	Amendments to the Food, Drug and Cosmetic Act of 1938 (United States Congress, 1962)
1966	—	Publication of U.S. Public Health Service regulations leading to creation of Institutional Review Boards for research involving humans (Levine, 1981)

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PMCID: PMC9196248
PMID: 35712451

On the Interplay Between the Medicine of Hildegard of Bingen and Modern Medicine: The Role of Estrogen Receptor as an Example of Biodynamic Interface for Studying the Chronic Disease's Complexity

Sabrina Melino[1,*,†] and Elisabetta Mormone[2,*,†]

Introduction

Hildegard of Bingen (1098-1179) interpreted the origins of chronic disease highlighting and anticipating, although only in a limited fashion, the importance that complex interactions among numerous genetic, internal milieu and external environmental factors have in determining the disease phenotype. Today, we recognize those factors, capable of mediating the transmission of messages between human body and environment and *vice versa*, as biodynamic interfaces.



I. Elements of Clinical Trials

A. What is the Question?/Protocol

- Each clinical trial must have a carefully selected, clearly defined primary question stated in advance.
- A standard format for the Study Protocol
- Example: TB Prevention (from web)

A RANDOMIZED, DOUBLE BLIND, PLACEBO CONTROLLED TRIAL TO DETERMINE THE EFFICACY OF ISONIAZID (INH) IN PREVENTING TUBERCULOSIS DISEASE AND LATENT TUBERCULOSIS INFECTION AMONG SOUTH AFRICAN INFANTS WITH PERINATAL EXPOSURE TO HIV

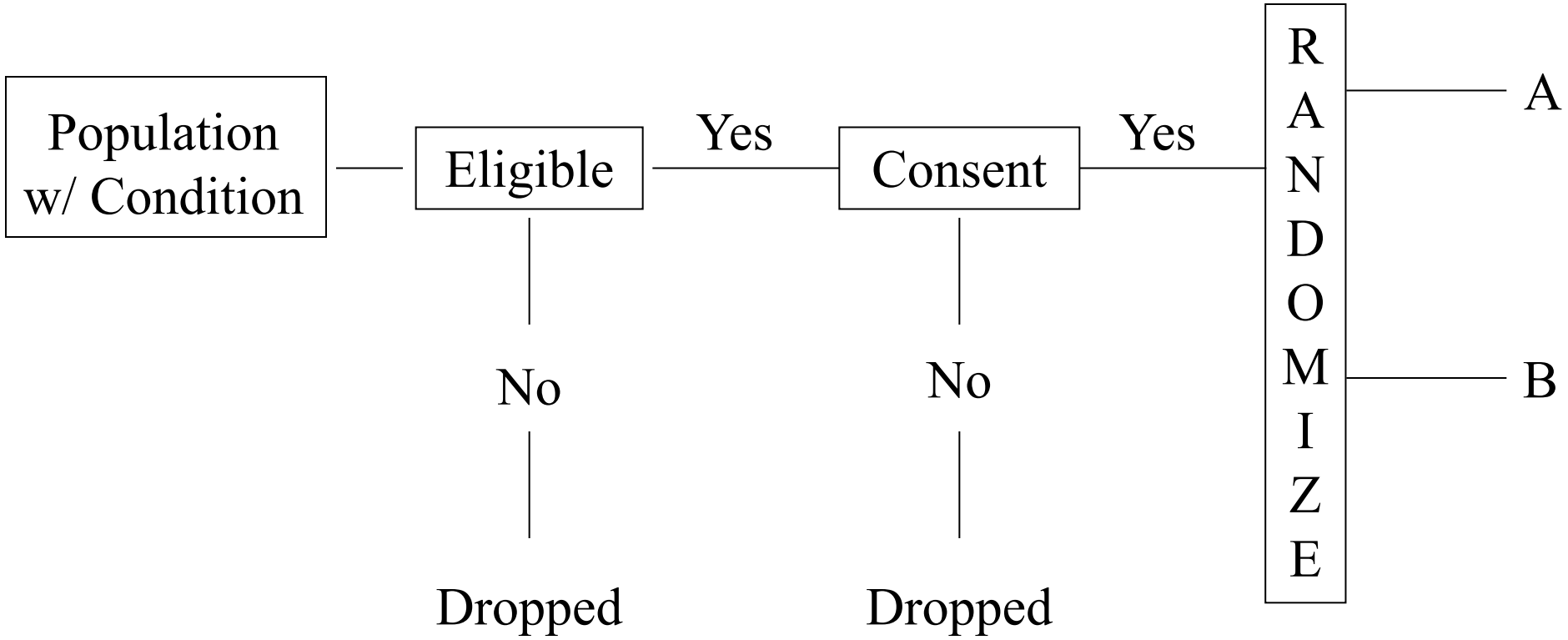
Sponsored by:

The United States of America National Institute of Allergy and Infectious Diseases
and
Secure the Future Foundation

https://www.impaactnetwork.org/studies/p1041

B. Basic Study Design:

Randomized Controlled Trial



Comment: Compare A with B

- Advantages
  - Randomization tends to produce comparable groups
  - Randomization produces valid statistical tests

- Disadvantage
  - Generalizable Results?
    Subjects may not represent general patient population: volunteer effect

  - Acceptability of Randomization Process
    Some physicians will refuse
    Some patients will refuse
    Ethical issues of randomization

Randomized Cross-Over Design

Scheme

		Period	
Group		I	II
AB	1	TRT A	TRT B
BA	2	TRT B	TRT A

- Advantage
  - Each patient acts as their own control
  - Smaller sample size
- Disadvantage
  - Not useful for acute disease
  - Disease must be stable
  - Assumes no period carry over

C. Masking or Blinding

Keeping the identity of treatment assignments masked for:

1. Subject
2. Investigator, treatment team, or clinical evaluator
3. Statistical evaluation team

Purpose of masking: bias reduction

Each group masked eliminates a different source of bias

Masking is most useful when there is a *subjective* component to treatment or evaluation

Types of Trials

- No Blind
  - All patients know treatment
- Single Blind
  - Patient does not know treatment
- Double Blind
  - Neither patient nor healthcare provider knows treatment
- Triple Blind
  - Patient, physician, and statistician/monitors do not know treatment
- Double blind recommended when possible

D. Other Issues

Sample Size: Clinical trials should have sufficient statistical power to detect differences between groups to be of clinical interest

Recruitment of Study Participants: One of the most difficult issues in a clinical trial is obtaining a sufficient number of participants in a reasonable period of time

Participant Adherence (Compliance): Measures are needed to monitor and enhance participant adherence

II. What Do the Data Show

A. Examples of Inference Based on Flawed Designs:

Internal Mammary Artery Ligation to Treat Angina Pectoris

Original study: Kitchell, Am. J. Cardiol. 1;46, 1958.

50 people treated then observed for 2-6 months:

- 18 patients reported no chest pain
- 16 patients reported fewer or less severe attacks
- 11 reported no improvement
- 5 patients died

Double Blind RCT: Cobb et al, NEJM, 260:1115, 1959.

No difference in stress test/ECG in the two groups

II. What Do the Data Show

Another Double-Blind Study with Sham Procedure for Control

Percutaneous coronary intervention in stable angina (ORBITA): a double-blind, randomised controlled trial

Methods ORBITA is a blinded, multicentre randomised trial of PCI versus a placebo procedure for angina relief that was done at five study sites in the UK. We enrolled patients with severe (≥70%) single-vessel stenoses. After enrolment, patients received 6 weeks of medication optimisation. Patients then had pre-randomisation assessments with cardiopulmonary exercise testing, symptom questionnaires, and dobutamine stress echocardiography. Patients were randomised 1:1 to undergo PCI or a placebo procedure by use of an automated online randomisation tool. After 6 weeks of follow-up, the assessments done before randomisation were repeated at the final assessment. The primary endpoint was difference in exercise time increment between groups. All analyses were based on the intention-to-treat principle and the study population contained all participants who underwent randomisation. This study is registered with ClinicalTrials.gov, number NCT02062593.

Interpretation In patients with medically treated angina and severe coronary stenosis, PCI did not increase exercise time by more than the effect of a placebo procedure. The efficacy of invasive procedures can be assessed with a placebo control, as is standard for pharmacotherapy.

Lancet 2018; 391: 31–40
Published Online November 2, 2017
http://dx.doi.org/10.1016/S0140-6736(17)32714-9

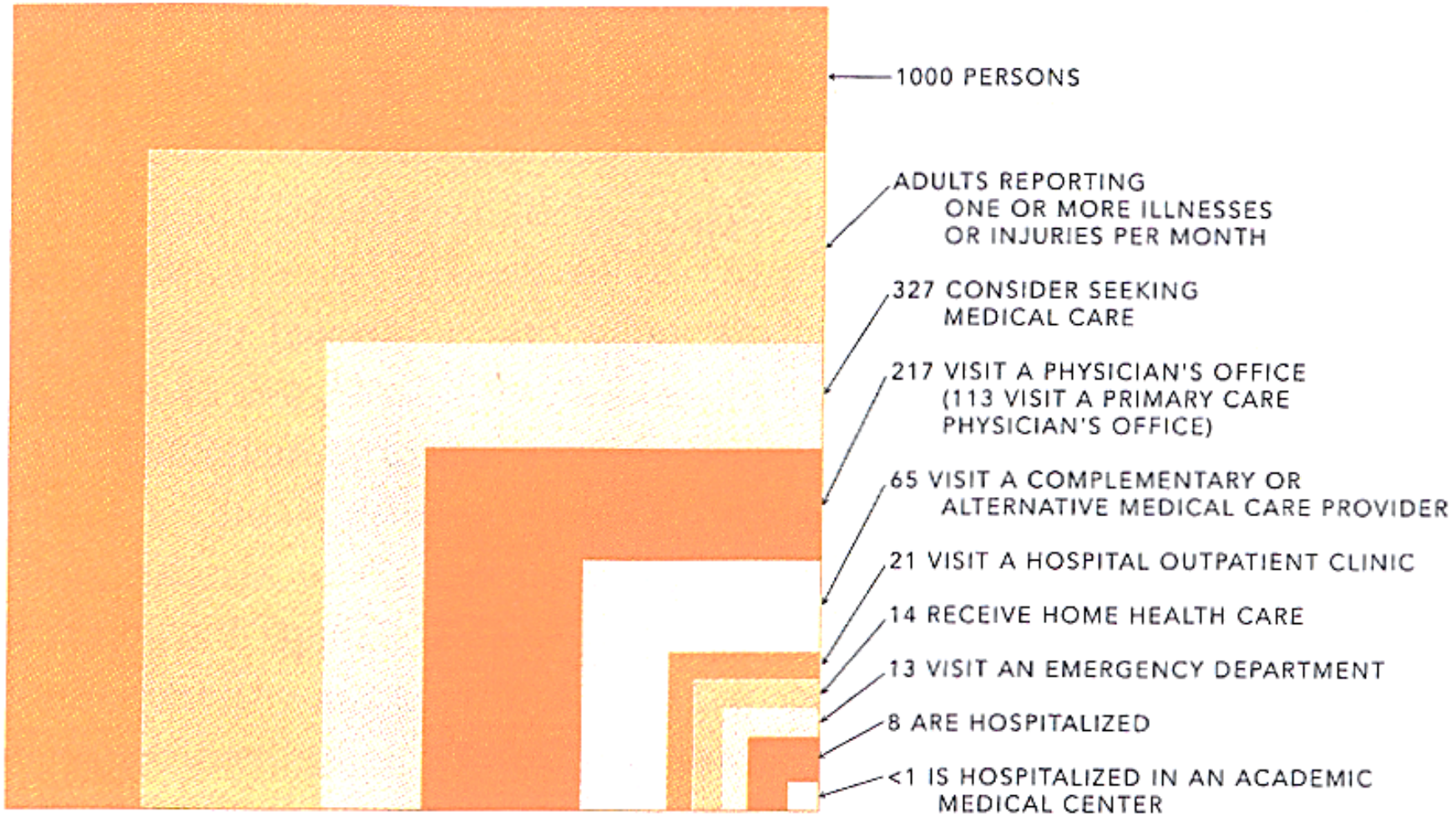


Figure 12–2 Estimates of the number of people in the United States who report illnesses and receive various forms of health care. Less than 1 in 1000 is hospitalized in an academic medical center. (*Redrawn from Fig. 2 of L. A. Green, et al., "The Ecology of Medical Care Revisited,"* N. Engl. J. Med. ***344:****2021–2025, 2001. Used by permission.*)